

Yan, D., Sadeghi, A., Yazdani-Asrami, M. and Song, W. (2024) Artificial Intelligence-driven model for resistive superconducting fault current limiter in future electric aircraft. *IEEE Transactions on Applied Superconductivity*, (doi: [10.1109/TASC.2024.3421903](https://doi.org/10.1109/TASC.2024.3421903))



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Deposited on: 09 July 2023

Artificial Intelligence-Driven Model for Resistive Superconducting Fault Current Limiter in Future Electric Aircraft

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Abstract—Fast and accurate electrothermal characterization of Superconducting Fault Current Limiters (SFCLs) is critically important for their performance evaluation. At the design stage of the SFCL, simulation is needed to validate the feasibility of the idea before fabrication. The most popular simulation models are the finite element models, thermal electrical analogy (TEA), equivalent circuit model and the finite difference method. However, all those four models are time consuming either in preparing stage or in running stage. In this study, we propose an Artificial Intelligence-driven model to achieve fast calculation and monitoring of SFCL. In this paper, a surrogate model was presented to estimate the resistance and temperature of SFCL, under different fault scenarios. These scenarios include different fault duration, voltages, maximum prospective fault current, and different operating temperatures. The TEA was used to electrothermally simulate the SFCL, under different fault scenarios. Then, temperatures of different layers in SFCL, as well as the total resistance of SFCL were collected. Afterwards, a Feed Forward Neural Network (FFNN) was used for SFCL characterization. The data of fault scenarios were fed into FFNN as inputs while temperature and resistance were considered as outputs. Also, to show the superiority of FFNNs, two other intelligent methods were used for surrogate modelling of SFCL. Results showed that FFNN had more than 98% accuracy for estimating different electrothermal parameters of SFCL. This intelligent surrogate model is a step forward towards developing digital twin for SFCL in future cryo-electric aircraft.

Index Terms—Artificial Neural Network, Characterization, Machine Learning, SFCL Design, Recovery Time.

I. INTRODUCTION

Electrified aircraft is a potential solution for the future aviation system, considering the emission standards set by the organizations and institutions in the UK, USA and Europe, such as Flightpath which demands a reduction of CO₂ emission by 75% in the year 2050 compared to a new aircraft baseline in the year 2000 [1, 2]. High Temperature Superconductor (HTS) devices were proposed for electrified aircraft systems to enhance the viability of cryo-electrification in aviation by offering high efficiency and large power density due to their ability of carrying high current [3, 4]. Superconducting Fault Current Limiters (SFCLs) are

protection solutions for electrical systems including the stand-alone power system of an electric aircraft [5].

When the fault occurs in an electric aircraft, high fault current may damage the equipment in the system and cause severe consequences [6]. Since, SFCLs limit the fault current, thereby, they protect other equipment from being damaged by the high short circuit current. Thus, online condition monitoring of SFCLs is challenging to keep the safety and reliability of power train. For condition monitoring, electrothermal characteristics of SFCLs should be estimated under different operational and fault conditions. On the other hand, to design an SFCL efficiently, simulation is needed before fabrication, and the conventional simulation models are time-consuming. This paper aims to demonstrate a novel model to improve the modelling and computational efficiency during the design process of SFCL.

Currently, Finite Element Method (FEM) [7-9], the Thermal Electrical Analogy (TEA) [10-12], Equivalent Circuit Model (ECM) [13, 14], and Finite Difference Method (FDM) [15-17] are used for electrothermal characterization of SFCLs. Although these modelling techniques are accurate, they require a huge amount of computational burden and time to characterize the SFCL.

FEM allows the geometrical based modelling, and the results can be very accurate if well-modelled [18]. However, the calculation speed of FEM is relatively slow, without the potential for being used in real-time modelling due to such slow computation burden. In addition, FEM is not flexible as it can be usually used during the design stage and not in the SFCL operation stage, which essentially means it is not a good option for any system-level study.

TEA can calculate the thermal profile that cannot be easily observed in experiment [19]. However, building a TEA model that takes into account the structural heat transfer of a complex system is time-consuming, and the simulation time will increase significantly. ECM is one of the fastest simulation models for modelling of superconducting devices [20], however, the ECM simplify the complex physical conditions into equivalent circuit, and this simplification can compromise accuracy, but it can be used for system-level modelling and study in electric aircraft. Similar to FEM, FDM can handle the cases with

Submitted 22 March 2024, revised 26 May 2024, accepted 27 June. This work was supported by the U.K. Engineering and Physical Sciences Research Council (EPSRC) under Grant EP/X525716/1. (Corresponding author: Wenjuan Song).

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irregular shape and boundary conditions [21], but also time-consuming. Each of the four methods exhibits promise within the realm of HTS fields; however, no singular approach can concurrently achieve timeliness, precision, and simplicity. However, they are impractical for online and real-time condition monitoring of SFCLs [22]. So, there is a need for fast and accurate models to estimate and predict the electrothermal characteristic of SFCLs, under different fault scenarios.

To achieve real-time models in electrified aircraft system, new simulation approaches with rapid computation and high reliability should be investigated such as Artificial Intelligence (AI) techniques [23, 24]. The AI methods have the ability of rapid processing and analysing of big data set and high precision of calculation. Although AI has been widely used in some engineering fields for many years, it is still a relatively new approach in the superconductivity community [25-30].

Although AI-based optimization algorithms have been recently used for design optimization of SFCLs [31-35]. Also, in [36], it is pointed out that the data-driven techniques have been used for estimating the characteristics of HTS tapes during fault. In [37-39], support vector machines were used to estimate the electrical parameters of SFCLs such as current, voltage, resistance, and the temperature of coolant fluid. The performed studies for estimating the electrothermal characteristic of SFCLs are limited to specific fault scenarios where operational conditions of SFCL such as voltage, prospective fault current, and temperature are constant while the structure of the SFCL is also considered to be unchanged. This has ended up with models for SFCLs that could be used just under specific design conditions and scenarios. Also, the conducted studies are more useful during the operation of SFCLs, where the SFCL has a fixed operational condition and a fixed structure. Thus, there is a gap in the literature for fast and accurate modelling of SFCLs where electrothermal characteristic of SFCL is estimated not only under different fault scenarios but also under changes of operational conditions and structure of SFCL that can be used during the design stage. Thus, such models could be flexible and adaptable with respect to any new changes in operational and structural condition of SFCL.

In this paper, an AI-driven surrogate model is proposed for a pancake-type SFCL, to bridge the existing gap. The proposed AI-based surrogate model can estimate the recovery time, the ratio of limited current to prospective fault current, maximum temperature across different layers, maximum fault current, maximum voltage across SFCL and maximum resistance of the SFCL. The development of the AI model involved an initial simulation phase using a TEA model for SFCL.

TEA model has been considered as a high accuracy model for simulating SFCL. In [12], TEA model has been used to simulate an BSCCO based SFCL, and there are just slight difference between simulation results and test results. In [11], the comparison between TEA model, FDM model and test results has been conducted, and the results show that the difference between TEA model and FDM model is very small. Moreover, both simulation models present high agreement with the experimental results, with a maximum 6% difference for the peak fault current.

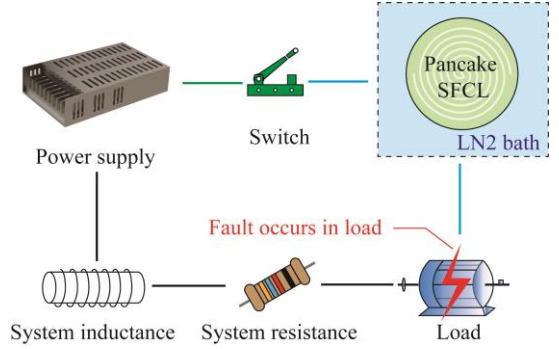


Fig. 1. Schematic of the fault circuit for SFCL

However, considering the large amount of data needed for the AI-driven model, using TEA model can significantly reduce the simulation time. The proposed TEA model in this paper has been validated by the experimental results by [40]. This TEA model was used to simulate SFCL under diverse conditions, generating a dataset comprising approximately 3000 sets of outcomes corresponding to different system states. We consider permutations with different fault cycle, voltage levels, prospective fault current, length of SFCL, module number and initial temperature. The data have then been trained by three different AI models, which are utilizing Feedforward Neural Networks (FFNN), Gaussian Process Regression (GPR), and Generalized Additive Models (GAM). The developed AI model, trained on this comprehensive dataset, demonstrates the capability to achieve high-accuracy simulations in remarkably short durations. This method enables efficient and accurate predictions for various SFCL scenarios. Having this AI-driven model for SFCL will pave the way for enabling researchers towards SFCL digital twin in future electric aircraft [41]. To date, we developed and published surrogate models for SFCL (in this paper), HTS cable [42], and superconducting magnet [43] in “Propulsion, Electrification & Superconductivity” group at University of Glasgow, UK.

II. SFCL MODELLING FOR DATA COLLECTION: METHODOLOGY AND STRUCTURE

A. SFCL simulation

The studied SFCL is implemented in a circuit consisting of an AC power supply, a resistor, an inductor, the load and the SFCL module which is shown in Fig. 1. The SFCL is in the liquid nitrogen (LN_2) bath. When the load is short-circuited, the current will increase significantly, thus the SFCL will start to quench and limit the current.

The HTS tape of the studied SFCL is a 10-layer tape, and the specification of the HTS tape is listed in Table 1 and its cross-section view is shown in Fig. 2. The HTS tape has two stainless-steel layers, two solder layers, two copper layers and two silver layers symmetrically placed on both sides of the tape. The YBCO layer and the Hastelloy layer are in the middle of the tape.

To be noticed, the thickness ratios in Fig. 2 are not the actual thicknesses but are just used as an indication. I_c is the critical current of the HTS tape and in the simulation model, we consider the I_c is homogeneously distributed over the length and thickness direction of the HTS tape. There are 9 turns in the coil where the HTS tapes are connected in series.

TABLE I. PHYSICAL SPECIFICATION OF THE HTS TAPE USED IN SFCL

Items	Values	Unit
Width	12	mm
Total thickness	0.283	mm
YBCO thickness	2	μm
Hastelloy thickness	50	μm
Stainless steel thickness	2*80	μm
Silver thickness	2*1.5	μm
Copper thickness	2*5	μm
Solder thickness	2*4	μm

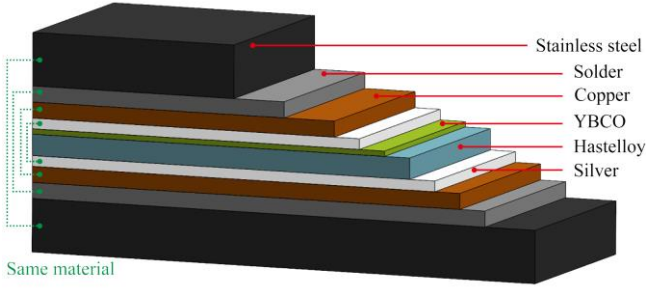


Fig. 2. Cross-section view of the HTS tape used for SFCL (layers are not drawn to the real dimensions.)

It worth mentioning that, in TEA modelling, the average temperature of the turns and the coil has been considered as the inputs for the AI model.

The TEA method has been applied to simulate the electrothermal characteristic of SFCL. Then, the results of simulations under different fault scenarios are collected for the AI-based model. The TEA-based simulation is conducted based on the Runge-Kutta method using MATLAB for solving the ordinary differential equations [11].

Fig. 3 shows the simulation flowchart. The simulation starts by defining the initial parameters of the system, including the system resistance, system inductance, initial temperature, initial conditions of different layers, number of SFCL modules, length of SFCL, and the current and voltage settings. After determining the initial conditions of the system, the simulation will start. The iteration will start by determining the system resistance. During normal condition (before fault), the resistance of the circuit is the summary of system resistance and the load resistance. During the fault stage, the resistance of the circuit is the summary of the SFCL resistance and system resistance. When the fault is cleared, the system resistance will be changed to $0.5 M\Omega$. After determining the system resistance, the current will be calculated. The resistance of the YBCO layer will be calculated based on the temperature results of the previous iteration. The resistivity calculation equation for YBCO layers is based on [44] and is shown in equation (1).

$$\rho = \frac{\left(\frac{E_0}{|J|^{n_1-1}} \right) \times \rho_{SCn}}{\left(\frac{E_0}{|J|^{n_1-1}} \right) + \rho_{SCn}} \quad (1)$$

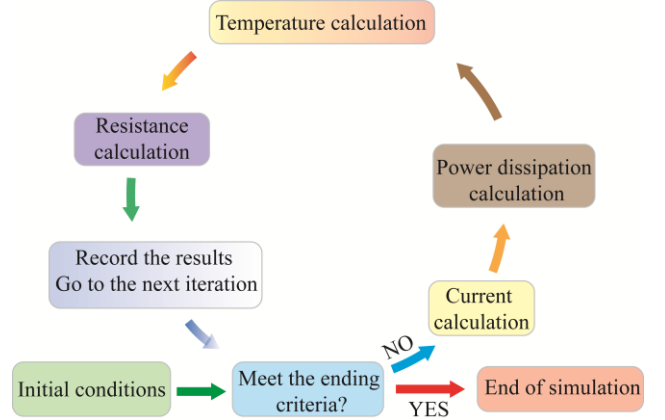


Fig. 3. Modelling flowchart of SFCL

$$J = \frac{I_{sp}}{a_{sp}} \quad (2)$$

$$J_c = \left(\frac{I_c}{a_{sp}} \right) \times \left(\frac{T_c - T_{sp}}{T_c - T_0} \right) \quad (3)$$

In which, ρ is the resistivity of the YBCO layer, $\rho_{SCn} = 9 \times 10^{-7} \Omega\text{cm}$ is a chosen fixed resistivity, which is the resistivity of the superconducting material in normal state [44], The $E_0 = 1 \times 10^{-6} \text{ V/cm}$ is the critical electric field strength, and the J is the current density.

The J_c is calculated by the temperature-dependent equation, which can be found in equation (3). The index value is $n_1 = 25$ and it depends on the superconducting material. The I_{sp} is the current of the YBCO layer, and the a_{sp} is the cross-section area of the YBCO layer. The T_{sp} is the temperature of the superconducting layer, the T_c is the critical temperature of YBCO and the T_0 is the temperature of the LN_2 .

Then, the power dissipation will be calculated to determine the temperature change of each layer. The convective heat transfer used for temperature calculation is determined by equation (4) [12]. In which, ΔT is the temperature difference between the stainless-steel layer and the LN_2 .

$$\begin{aligned} & \text{if } \Delta T < 2 \quad h = 0.18 \\ & \text{if } 2 < \Delta T < 26.5 \quad h = \frac{[a_0 + a_1 \Delta T + a_2 \Delta T^2 + a_3 \Delta T^3 + a_4 \Delta T^4 + a_5 \Delta T^5]}{\Delta T} \\ & \text{if } \Delta T > 26.5 \quad h = 0.03 \end{aligned} \quad (4)$$

where:

$$\begin{aligned} a_0 &= 3.8147, a_1 = -2.8209, a_2 = 0.72328, \\ a_3 &= -0.049438, a_4 = 1.325 \times 10^{-3}, a_5 = -12.773 \times 10^{-6} \end{aligned}$$

After calculating the temperature of each layer, the resistance of different layers will be calculated. Then, the simulation will stop if the stoppage criteria are met, if it doesn't, it will go to the next iteration. Two different ending criteria have been utilized for this model, and the simulation will be stopped when either of these two criteria is met. The first trigger condition is related to recovery, when the temperature of the superconducting layer is reduced to 110% of the initial temperature after the fault is cleared, we consider that the SFCL has been fully recovered, so we set this temperature point as the first trigger condition. The second ending criterion is related to

TABLE II. SWEEP PARAMETERS FOR THE INPUTS OF THE AI-DRIVEN MODEL

Inputs	Values
Fault cycles	2/ 4/ 5/ 8/ 10/ 12
Voltage	115 V/ 230 V/ 400 V/ 600 V/ 800 V/ 1000 V
Prospective fault current	10 kA/ 15 kA/ 20kA
Fault energy	$Voltage * Prospective\ fault\ current$
Length of SFCL	10 m/15 m/30 m
Module number	1/2/3
Initial temperature	77 K/70 K/65 K
Critical current	533 A/826 A/1023 A

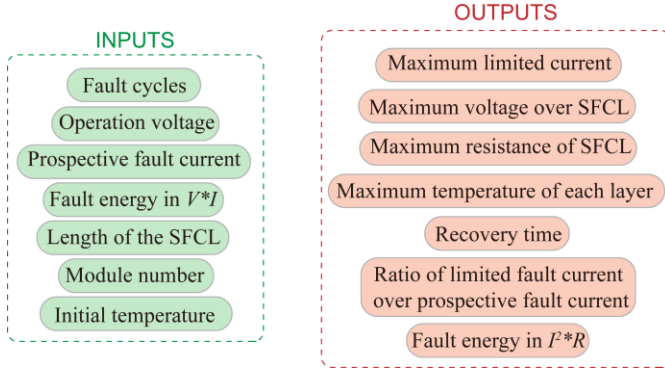


Fig. 4. The list of inputs and outputs data collected for AI-based surrogate model

the temperature limit of the HTS tape, when the temperature of the YBCO layer reaches 600 K, we consider that the HTS tape will burn and thus the simulation will end [45, 46].

The inputs and outputs for the AI model are shown in Fig. 4 and Table 2. We consider the fault cycles from 2 to 12, and the voltage from 115 V to 1000 V. The prospective fault current is set as 10 kA, 15 kA and 20 kA. The fault energy for the input can be calculated by $V \times I_{\text{prospective fault}}$. The length of the SFCL is set as 10m, 15 m and 30 m. The module number of SFCL is set as 1, 2 and 3. The initial temperatures are 77 K, 70 K and 65 K. The considered critical currents are 533 A, 826 A and 1023 A accordingly. All the cases are under 50 Hz.

Based on the current number of inputs, our dataset has approximately 3000 cases. The outputs for the AI model include the maximum limited current, the maximum voltage over SFCL, the maximum resistance of SFCL, the maximum temperature of each layer, the recovery time, the ratio of the limited current over prospective fault current and the maximum fault energy (calculated by $I^2 \times R$, in which, I is the current of each layer and R is the resistance of each layer) during each simulation.

The simulation results under different fault durations are shown in Fig. 5. The solid black line indicates the results under 5 fault cycles (100 ms) while the red dot line indicates the results under 2 fault cycles (40 ms). Fig. 5(a) shows the current limiting profile, and the current has been limited from 10 kA to 1434 A under both cases during the first cycle and has been further limited in the following cycles. However, the current was limited to 702 A when the fault was cleared under the 40 ms case, while the current was limited to 481 A when the fault was cleared under 100 ms case.

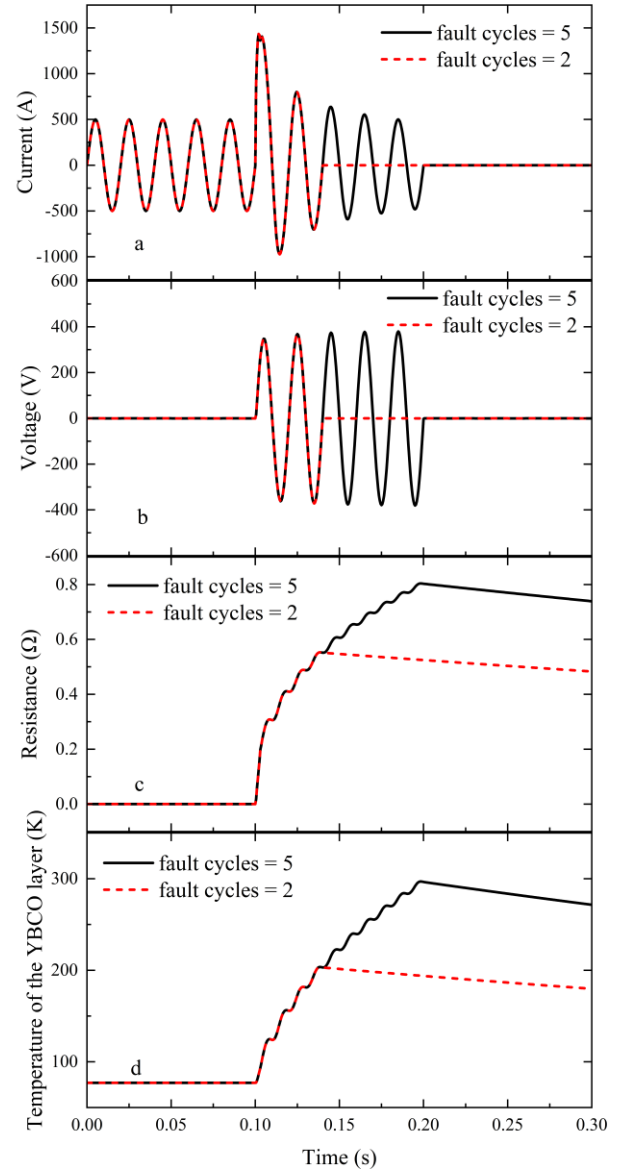


Fig. 5. Results of the different cases under 2 and 5 fault cycles in current, voltage, resistance, and temperature

The voltage profile is shown in Fig. 5(b). It can be found that the voltage over the SFCL is almost 0 V before fault, due to the zero-resistance characteristics of the HTS element. However, after the fault happens, the SFCL starts to quench and the resistance increases rapidly, as can be seen in Fig. 5(c), and consequently, the voltage over SFCL increases. The voltage profile shows that the voltage taken on SFCL under two different cases are the same during the first two cycles, and the voltage of the 40 ms case drop to 0 first after fault cleared.

Fig. 5(c) shows the resistance profile under different fault cycles, and according to the figure, the resistance of the 100 ms case reaches 0.8 Ω before the fault is cleared, while the resistance of the 40 ms case reaches to 0.5 Ω before the fault was cleared.

The resistance of the SFCL is highly dependent on the temperature, and according to Fig. 5(d), the temperature profile of the YBCO layer shows the same trend as the resistance profile, the temperature of the 100 ms case reaches

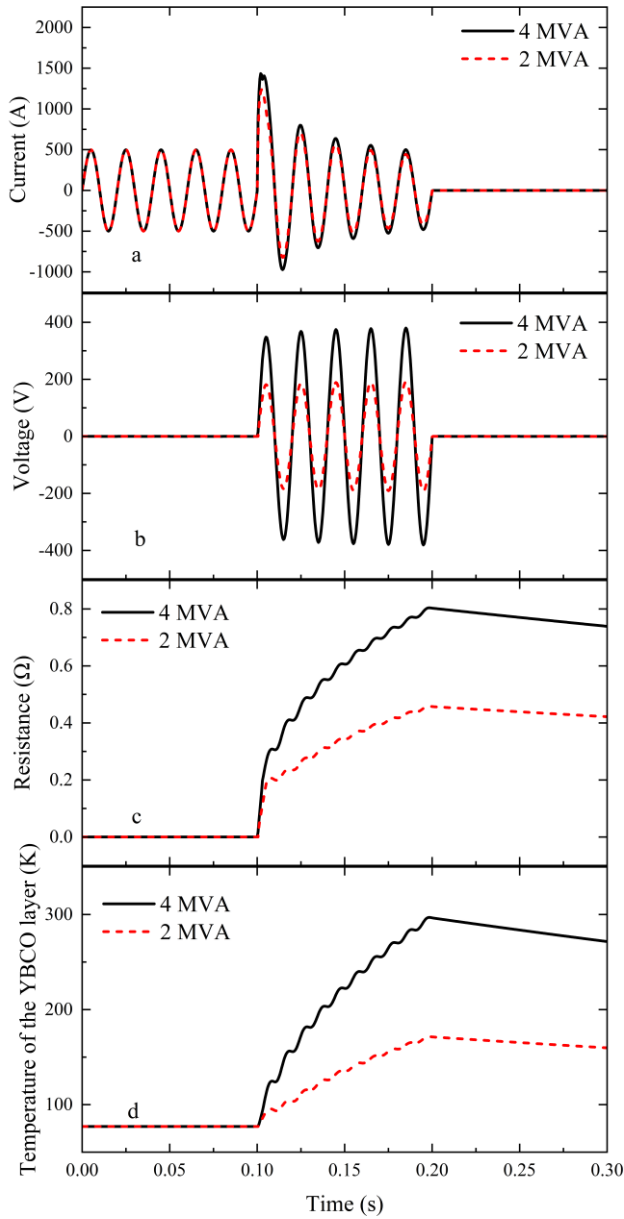


Fig. 6. Results of the different cases under 2 MW and 4 MW in current, voltage, resistance, and temperature

297 K before the clearance of the fault, while that of the 40 ms case reaches to 203 K before the clearance of fault.

Fig. 6 shows the current, voltage, resistance and temperature profile of the SFCL under different fault energy levels. According to Fig. 6(a), the current profile is the same under both cases before fault, and after the fault happens, the SFCL limit the current from 10 kA to 1434 A and 1239 A in the first cycle respectively under 4 MVA and 2 MVA.

Fig. 6(b) shows the voltage profile of two cases, and in this figure, the voltage taken under the 4 MVA case reaches 348 V when fault happens, and gradually increases to 380 V before the fault clearance. Similarly, the voltage taken under the 2 MVA case reaches 181 V when fault occurs and increases to 191 V in the fifth cycle before the fault clearance. The resistance profile and the temperature profile are shown in Fig. 6(c) and (d), the maximum resistance and temperature under the 4 MVA case

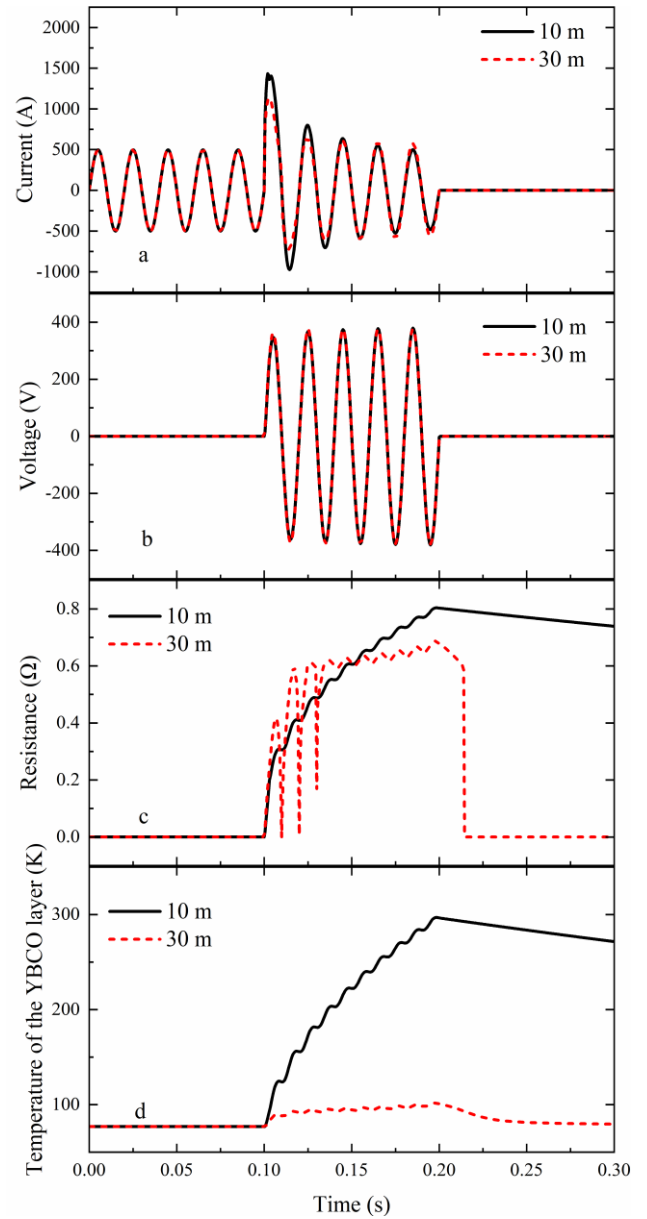


Fig. 7. Results of the different cases with 10 m and 30 m length HTS tape in current, voltage, resistance, and temperature

are 0.8 Ω and 297 K respectively, while that of the 2 MVA case are 0.45 Ω and 171 K.

Fig. 7 shows the current, voltage, resistance and temperature profile of the SFCL with different HTS tape lengths. According to Fig. 7(a), the maximum fault current of the 10 m HTS tape case is limited to 1434 A, while it is limited to 1135 A for the 30 m HTS tape case. The voltage profile of the two cases is shown in Fig. 7(b), and the voltage over SFCL increases to 348 V and 362 V for 10 m and 30 m HTS tape cases respectively. The voltage over SFCL reaches 380 V and 377 V respectively during the last fault cycle.

Fig. 7(c) shows the resistance profile of two cases, it can be found that the resistance of the 10 m HTS tape case is lower at the beginning, and higher after 0.15 s. The maximum resistance of the 10 m and 30 m HTS tape cases are 0.8 Ω and 0.7 Ω respectively. The temperature profile is shown in Fig. 7(d),

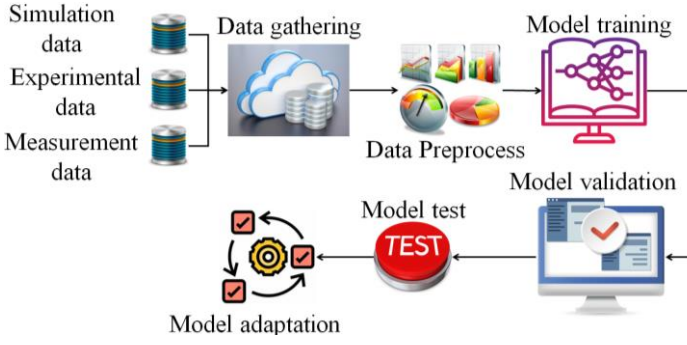


Fig. 8. The general procedure of implementing AI-based models for regression purposes

the maximum temperature of the 10 m and 30 m HTS tape cases are 297 K and 101 K respectively. The resistance of the 10 m HTS tape case is lower at the beginning due to the shorter length, however, during the fault, the temperature of the 10 m HTS tape case is much higher than the 30 m case, hence the increasing speed of the resistance under 10 m case is faster. Due to the asynchronous increase resistance of two cases, the limited current of the 10 m HTS tape case is higher at the beginning of fault, but lower at the end.

III. AI-based Modelling Methodology

A. Introduction of AI techniques

AI-based techniques could be used for different purposes such as design/parameter optimization, regression, classification, clustering, and decision-making [27]. In this context, the regression task is referred to modelling the relationship between the inputs of the system and the outputs of the model. Regardless of the objective of AI-based techniques, they usually consist of six stages, data collection, data preprocessing, training, validation, testing, and model adaptation.

As shown in Fig. 8, the initial stage is data collection where these data could be originated from experimental tests, simulations, real-time data gathered by sensors, or a combination of them. At this stage, the inputs, and outputs of the AI-based model should be selected. The inputs are referred to the features, variables, or attributes that AI-based technique uses to characterize the output. Inputs could be independent from each other or could have interdependencies. On the other hand, outputs are defined as the characterization of the system, that AI-based technique aims to predict [28]. Then, there is the data preprocessing step, where this step could include different levels such as data cleaning, data transformation, data encoding, imbalanced data handling, and feature extraction. More discussions about these steps and how to conduct them, are discussed in [47].

Following the data preprocessing step, data are divided into three subsets, train data, validation data, and test data. Afterwards, the training step starts where the AI-based model updates its properties to minimize the difference between estimated train outputs (ETO) and expected train outputs (XTO). At this stage, if the difference between ETO and XTO met the stoppage criteria, usually shown by ϵ , the model will proceed to the next stage [22]. Train stoppage criteria could be different indices, such as the maximum number of epochs, minimum value of the difference between ETO and XTO, etc.

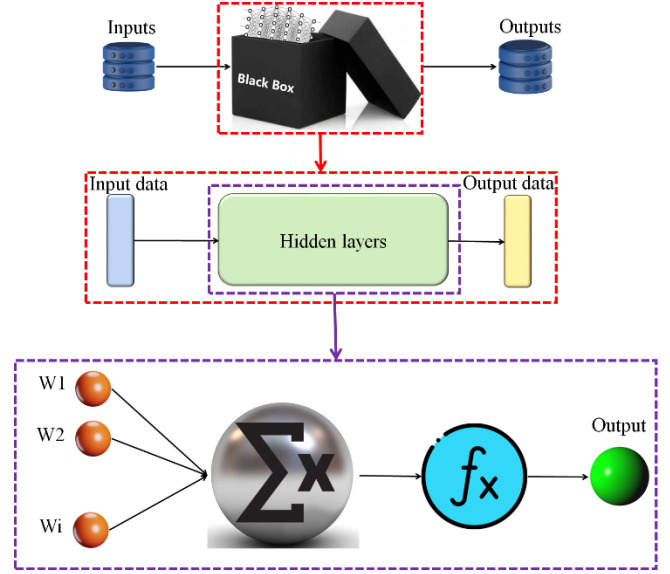


Fig. 9. A representative figure of surrogate model based on FFNN

In this regard, the epoch is defined as one complete pass by AI-based technique through the entire training dataset during the training phase while AI-based technique updates itself. When the training stoppage criteria have been met, the validation stage starts [22]. At this stage, the unseen validation dataset is fed to the trained AI-based model. If the trained model is precise enough to accurately predict the validation data, the test stage starts. On the other hand, if the trained model fails to show a high accuracy, the validation stops, and again training process starts. Just like the validation stage, in the test stage, unseen data are fed into the trained and validated model to test its accuracy [22]. The difference between the test and validation stages is that unlike the validation stage, if the trained model fails to accurately estimate the test dataset, there is no return to the training. Finally, there is the adaptation stage which is not the case for all AI-based models. Adaptation is just applicable for online/real-time models that data is constantly flowing into the model.

B. Feed Forward Neural Network

Feed Forward Neural Network (FFNN) is a type of neural networks that consists of units acting as “multiple inputs/single output”. This means that each unit of FFNN has multiple inputs, usually from previous layers, while it has just one output. Each one of these units called “neurons”. All layers, rather than main inputs and main outputs, are called hidden layers. The term “hidden” here means that during the training process, the values and operations happening in these layers are not directly observable. Assuming that the number of hidden layers in an arbitrary FFNN is (N_h) , the total number of layers in the FFNN would be $(N_h + 1)$. A typical FFNN is shown in Fig. 9, where a unit/neuron is connected to the adjacent unit/neuron through an assigned weight, that is updated during the training procedure of FFNN. Equations (5) and (6) show the received input of the i^{th} neuron of the p^{th} layer[43]:

$$U_j^p = \sum_{i=1}^{n_{p-1}} w_{ji}^{p-1} \cdot O_i^{p-1} + \theta_j^p \quad (5)$$

$$O_j^p = f(U_j^p) \quad (6)$$

where, U_j^p is the input value to the activation function, n_{p-1} is the number of neurons in $(p-1)^{th}$ layer, w_{ji}^{p-1} is the weight of the connection between i^{th} unit of $(p-1)^{th}$ layer and j^{th} unit of p^{th} layer, O_j^p is the output of the i^{th} unit in the $(p-1)^{th}$ layer, θ_j^p is the bias factor of the j^{th} unit in the p^{th} layer, and f is the activation function. Multiple activation functions could be considered to make a logical relation between inputs and outputs of neurons, which are discussed in detail by [48, 49]. For training the FFNN and updating the bias factors as well as the weights, usually, the Back Propagation Gradient Descent (BPGD) algorithm is used. In this method, the weights and bias factors are iteratively updated so that the objective function shown in equation (7) is minimized. The aim of this objective function is to reduce the error between ETO and XTO [50, 51].

$$E = \frac{1}{2} \sum_{q=1}^{n_p} \sum_{j=1}^{n_N} (O_j^{q,N} - T_j^{q,N})^2 \quad (7)$$

where, $O_j^{q,N}$ is the output value of the j^{th} unit in the output at the N^{th} layer for the q^{th} training iteration and $T_j^{q,N}$ is the output value of the j^{th} unit in the output at the N^{th} layer for the q^{th} training iteration.

Usually, the BPGD algorithm starts with randomly generation of weights and bias factors with small values. Then, forward propagation starts, where the input data, from input layers, propagate through hidden layers to reach the output values, aiming to minimize the objective function of equation (7). Finally, back propagation initiated after evaluation of (E) for all train inputs, the aim of back propagation is to update the weights and bias factors. Bias factors and weights are updated in back propagation, based on equations (8) to (11) [50, 51].

$$\Delta w_{ji}^{q-1,p-1} = -\frac{\partial E}{\partial w_{ji}^{p-1}} \quad (8)$$

$$w_{ji}^{q,p-1} = w_{ji}^{q-1,p-1} + \alpha \cdot \Delta w_{ji}^{q-1,p-1} \quad (9)$$

$$\Delta \theta_j^{q-1,p} = -\frac{\partial E}{\partial \theta_j^p} \quad (10)$$

$$\theta_j^{q,p} = \theta_j^{q-1,p} + \beta \cdot \Delta \theta_j^{q-1,p} \quad (11)$$

This procedure continues until the stoppage criteria are met, which could be the maximum number of iterations, maximum number of epochs, and/or minimum value for (E).

C. Gaussian Process Regression

Gaussian Process Regression (GPR) is a type of probabilistic model that operates based on kernel functions. Here, the kernel is any function that models the relationships between data points. Assuming that the dataset is represented as $\{(x_i, y_i); i = 1, 2, \dots, n\}$, it needs to be determined through a GPR-based model. In this dataset both inputs (x_i) and outputs (y_i) have unknown distributions. The GPR-based models aim to explain outputs (y_i) through the introduction of latent variable ($l(x_i)$), so that by implementing the latent variable, joint distribution of $l(x_i)$'s is Gaussian while explicating the basis function b . In this context, latent variables are those ones that are not directly

observed but are inferred from the observed data. Then, the covariance function is used for latent variables and project inputs into the output space with p dimension [52].

The definition of Gaussian process is conducted through two basic statistical functions, mean and covariance. By considering $m(x) = E(l(x))$ as the mean, and $k(x, x') = Cov[l(x), l(x')]$ be the covariance function, the GPR model is defined as equations (12) to (14) [53].

$$y = b(x)^T \beta + l(x) \quad (12)$$

$$l(x) \sim GP(0, k(x, x')) \quad (13)$$

$$b(x) \in \mathbb{R}^p \quad (14)$$

$k(x, x')$ is usually estimated as a hyperparameter, known as θ that shows the linearity level of the Gaussian Process model, more information about this can be found in [54].

Generally, to estimate the hyperparameters of the GPR model such as β , θ , logarithmic likelihood marginal function is used, as it is shown in equation (15). This equation is to be minimized by considering the $K(X, X|\theta)$ to be the covariance function/kernel function matrix, as it is shown in equation (16). The algorithm obtains the likelihood function of equation (16), then likelihood profiled of hyperparameters must be maximized [53].

$$\left[\begin{pmatrix} k(x_1, x_1) & \cdots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \cdots & k(x_n, x_n) \end{pmatrix} \right] \quad (15)$$

$$\begin{aligned} \log P(y|X, \beta, \theta, \sigma^2) = & -\frac{1}{2} \{ (y - B\beta)^T [K(X, X|\theta) + \sigma^2 I_n]^{-1} (y - B\beta) \} - \frac{n}{2} \log 2\pi \\ & - \frac{1}{2} \log |K(X, X|\theta) + \sigma^2 I_n| \end{aligned} \quad (16)$$

D. Generalized Additive Model

For conceiving the Generalized Additive Model (GAM), Generalized Linear Regression (GLR) should be introduced first. In GLR, output variables of Y_i are estimated by the input vector of X_i by considering the probability of $Pr(Y_i = y_i | x_{i1}, x_{i2}, \dots, x_{ip})$. The Pr is a probability function with a normal distribution of variance σ^2 and mean value of η_i which is defined as equation (17) [55]:

$$Pr(Y_i = y_i | x_{i1}, x_{i2}, \dots, x_{ip}) \sim N(\sigma^2, \eta_i) \quad (17)$$

Then, the output of the GLR is a weighted sum of the p predictors, as shown in equation (18) [15]:

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} \quad (18)$$

GAM uses the same procedure of GLR with a small difference that is known as non-parametric solution, as shown in (19) [55]:

$$\eta_i = \underbrace{\beta_0 + \beta_1 x_{i1}}_{\text{Parametric part}} + \underbrace{f_1(x_{i2}) + f_2(x_{i3}, x_{i4})}_{\text{non-parametric part}} \quad (19)$$

Here, the non-parametric part considers the unknown functions instead of using simple polynomial functions. Usually, these two functions are considered as smoothing splines. A

smoothing spline is a type of fitting technique for smoothing a curve to a set of data points [56].

E. Performance Indices

To analyze the accuracy and preciseness of the AI-based techniques, multiple indices have been used in this paper which are also commonly used in literature. The indices used in this paper include Absolut Error (AE), shown in equation (20), Relative Error (RE), shown in equation (21), Root Mean Squared Error (RMSE), shown in equation (22), goodness of fit (R), shown in equation (23), Mean Absolute Percentage Error (MAPE) shown in equation (24), and Residual Value (RV) shown in equation (25) [57-61]:

$$AE = |o_i - f_i| \quad (20)$$

$$RE = \left| \frac{o_i - f_i}{o_i} \right| \times 100\% \quad (21)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n_{data}} (f_i - o_i)^2}{n_{data}}} \quad (22)$$

$$R = \frac{\sum_{i=1}^{n_{data}} (o_i - \bar{o})(f_i - \bar{f})}{\sqrt{\sum_{i=1}^{n_{data}} (o_i - \bar{o})^2 \sum_{i=1}^{n_{data}} (f_i - \bar{f})^2}} \quad (23)$$

$$MAPE = \frac{1}{n_{data}} \sum_{i=1}^{n_{data}} \left| \frac{o_i - f_i}{o_i} \right| \times 100\% \quad (24)$$

$$RV = 1 - R \quad (25)$$

In equations (20) to (25), o_i is the target output, f_i is the estimated output, n_{data} is the number of data, $\bar{f}$ is the mean value of estimated outputs, and $\bar{o}$ is the mean value of target outputs.

IV. RESULTS AND ANALYSIS

The dynamic characteristics of the SFCL, such as the current limitation profile, voltage across the SFCL, resistance profile, and temperature profile, have been considered during the implementation of AI-based techniques. The proposed model considers several different approaches, utilizing FFNN, GPR, and GAM to capture the intricate behaviour of SFCLs under various operational conditions.

A. Sensitivity Analysis on hyperparameters of FFNN

The aim of sensitivity analysis is to select the hyperparameters of FFNN so that it results in the highest accuracy, lowest error, and lowest computational time. In other terms, for selecting the best hyperparameters, a compromise between accuracy and simulation time is needed. The first hyperparameters to be studied are the number of hidden layers and neurons. Fig. 10(a) shows the impact of increasing the number of hidden layers and neurons, on the RMSE value of the model.

TABLE III. IMPACT OF TRAINING METHODS ON THE ACCURACY OF THE SURROGATE FFNN MODEL

Training method	R value	RMSE	NDD (%)	Train Time (s)
Levenberg-Marquardt backpropagation	0.99999	0.4450	0.1236	250
Bayesian Regulation backpropagation	0.99999	0.7291	0.1621	296
BFGS quasi-Newton backpropagation	0.99491	15.1229	66.5904	43
Conjugate gradient backpropagation with Fletcher-Reeves updates	0.99956	5.2048	29.7483	44
One step secant backpropagation	0.99986	2.8729	16.2471	38

The RMSE reduces by increasing the number of hidden layers and neurons. As can be seen from the figure, the best performance of the model is typically achieved with three hidden layers, with minor changes made after increasing the number of neurons to more than 7. The same trend can be seen in Fig. 10(b) which shows the goodness of fit for different values of hidden layers and neuron numbers. The R-value of the model is highest for three hidden layers while it experiences minor changes for neurons more than 7.

Fig. 10(c) shows the percentage of data with accuracy lower than 2%, known as NDD. NDD reduces with the increase in the number of hidden layers and neurons where NDD shows the importance of increasing the number of neurons.

So far, Figs 10(a) and 10(b) approved that 3 hidden layers is the best structure for the proposed FFNN-based model. NDD, shown in Fig. 10(c), illustrates that when number of hidden neurons is 5, about 30% of data have accuracy higher than 2%. By increasing the number of neurons to 7, NDD reduces to 11%, which means 63% reduction. By further increasing the neuron number to 9, NDD would be reduced to approximately 0%. This means that by selecting the number of neurons to be 9, most of the data have more than 98% accuracy.

Fig. 10(d) demonstrates the training time changes based on the structure variations of FFNN. As shown in this figure, the selected structure, 3 hidden layers with 9 neurons, has about 250 seconds of training time.

After the selection of the number of layers and neurons, the sensitivity analysis is conducted on the training methods used for updating the weights and bias factors of the FFNN model. The results are tabulated in Table 3 where five different training methods have been used. Best performance, regarding the R values, RMSE, and NDD belongs to Levenberg-Marquardt backpropagation and Bayesian Regulation backpropagation.

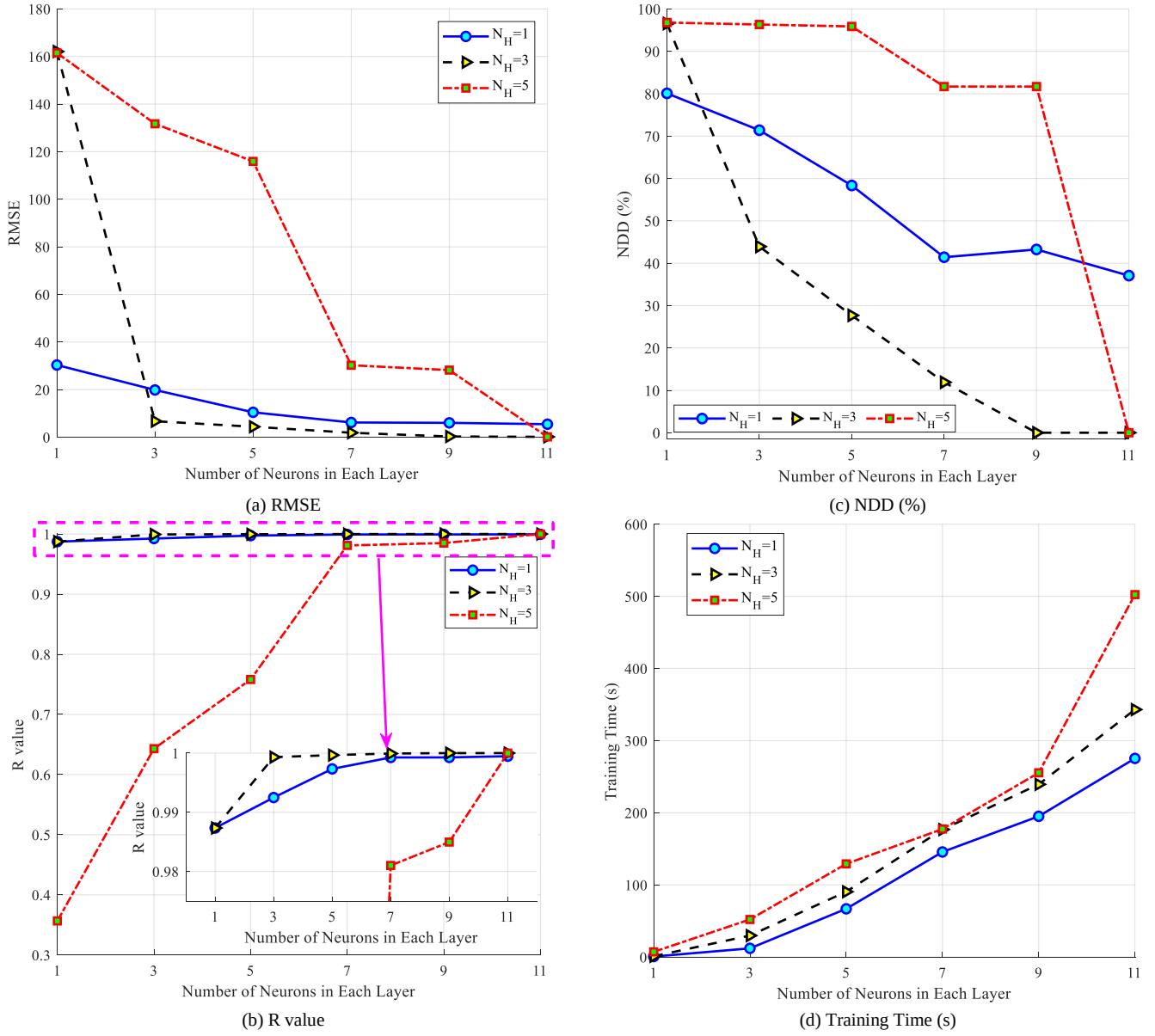


Fig. 10. Sensitivity analysis of FFNN

TABLE IV. IMPACT OF ACTIVATION FUNCTIONS ON THE ACCURACY OF THE FFNN SURROGATE MODEL

Activation Function	R value	RV	RMSE	NDD (%)	Train Time (s)
Tangent sigmoid	0.99999	1.1 e-9	0.4450	0.1236	250
Log-sigmoid	0.99999	1 e-7	0.8245	1.3730	223
Linear	0.99999	1.6 e-8	0.5435	0.1190	272
Saturating linear	0.99999	1.2 e-9	0.4511	0.1922	274
Positive linear	0.99999	1.5 e-8	0.5148	0.1623	274

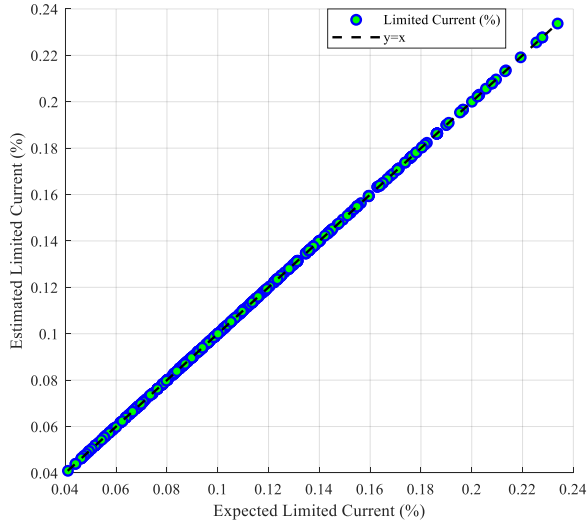


Fig. 11. The regression plot for the expected current limiting ratio versus the estimated current limiting ratio.

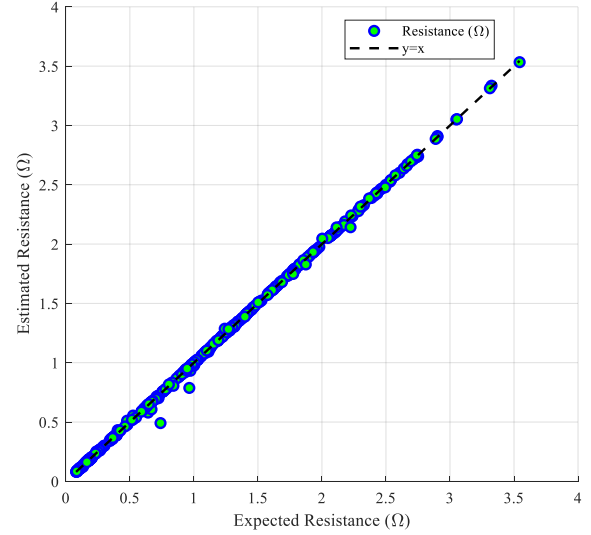


Fig. 13. The regression plot for the expected resistance versus the estimated resistance.

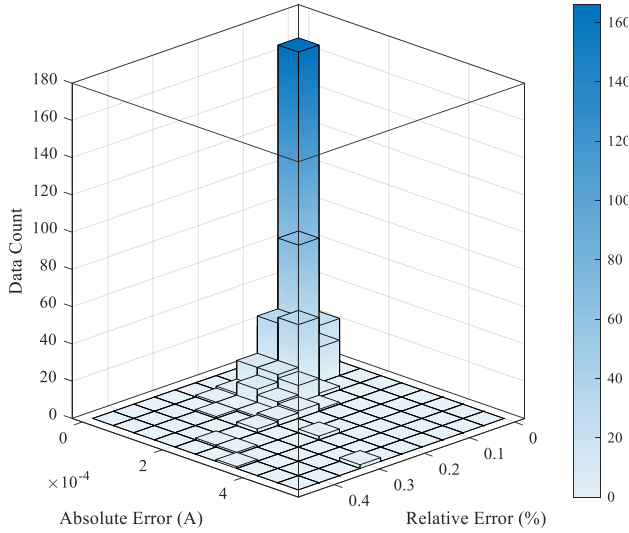


Fig. 12. Absolute error & relative error of estimation of the current limiting ratio.

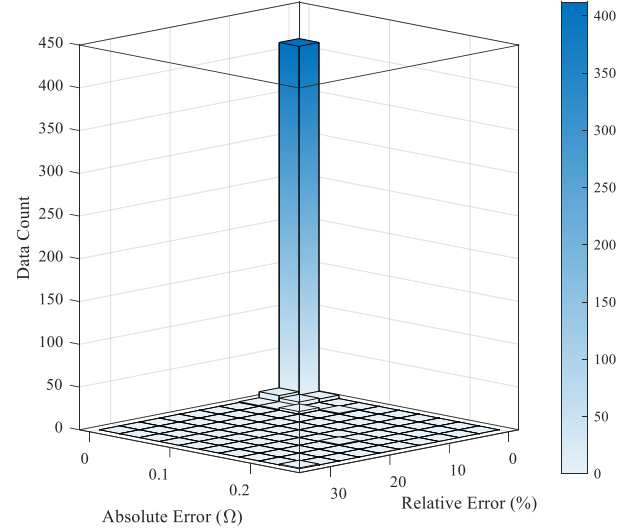


Fig. 14. Absolute error & relative error of estimation of the resistance of SFCL.

These two methods have the lowest NDD and RMSE and highest R values. However, their training time is 10x higher than other training methods. Between these two methods, Levenberg-Marquardt backpropagation has better performance, which is selected as the training method for the FFNN surrogate model. Table 4 tabulates the impact of the type of activation functions on the accuracy of the FFNN model. As shown in this table, the best performance is related to tangent sigmoid with the lowest NDD and RMSE and highest R value. Although the saturating linear function has also a close accuracy to the tangent sigmoid function, due to its higher training time, it is not considered as activation function for the FFNN surrogate model. It should be noted that the RV of the “tangent sigmoid” and “saturating linear” activation functions is about 10x to 100x lower than other activation functions which shows their high accuracy.

B. Results for best hyperparameter selected FFNN

Current limiting ability is among the most important factors of SFCL performance.

In this study, the current limiting ratio (the limited current over prospective fault current) is considered to indicate the current limiting ability of the SFCL. Fig. 11 illustrates the expected limited current versus the estimated one where the accuracy of the regression could be measured by the $y=x$ line. This meant that the more accurate the FFNN-based model is, the blue line behaves closely to the $y=x$ line. This means that the estimated values of the limited current are in excellent coordination with expected outputs, which indicates a highly accurate regression. Fig. 12 shows the relative error and absolute error of the current limiting ratio. Fig. 12 is a three-dimensional histogram delineating the distribution of data points based on their absolute error and relative error, with the z-axis representing the count of data points for each error category. The absolute error is plotted on x-axis and the relative error is plotted on y-axis. There is a significant data points concentration in the region corresponding to low absolute error and low relative error. This concentration indicates a high precision of the FFNN model in current limiting predictions.

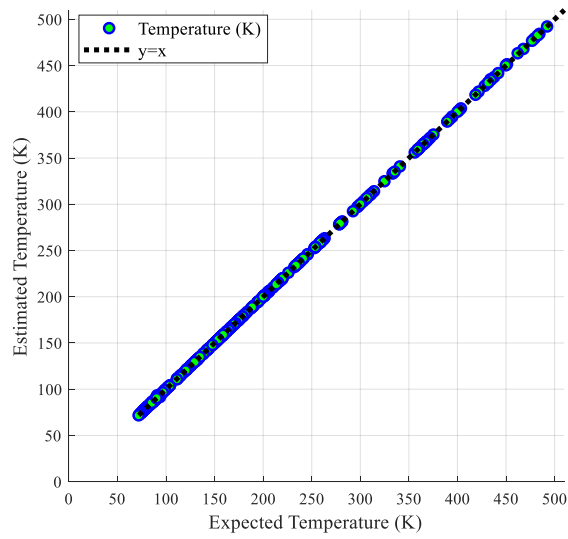


Fig. 15. The regression plot for the expected temperature of superconducting layer versus the estimated temperature of superconducting layer.

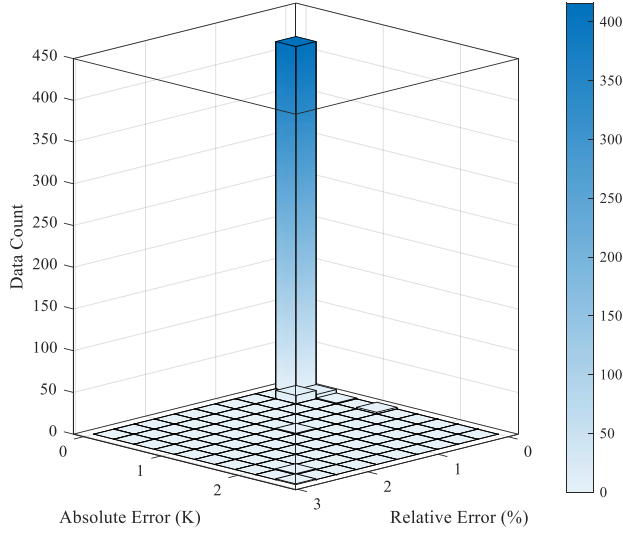


Fig. 16. Absolute error & relative error of estimation of the temperature of superconducting layer.

Fig. 13 demonstrates the goodness of fit for the estimation of the total resistance of the SFCL. As shown in this figure, except for 4 points, the results of FFNN-based estimation are in excellent coordination with the expected resistances that resulted in R-value of 0.9998. This is approved by observing that most of the blue circles behave closely with the $y=x$ line, as an accuracy indicator of the model.

The absolute error and relative error of the SFCL resistance estimation are presented in Fig. 14, where more than 99% of the data have the absolute error of 0 Ohm and the relative error of 0 %. This means that 99% of the resistances have been estimated with high accuracy by FFNN. In other words, Fig. 14 presents the absolute error of resistance estimation in ohms (Ω) against relative error in percentage (%) on the horizontal plane, while the vertical axis reflects the data count.

The absolute error scale on the x-axis ranges from 0 to over 0.3 Ω , and the Relative Error scale on the y-axis spans from 0 to 30%. Noteworthy, there is a peak at the lower end of both error axis, and the tallest bar indicating that most of the data points

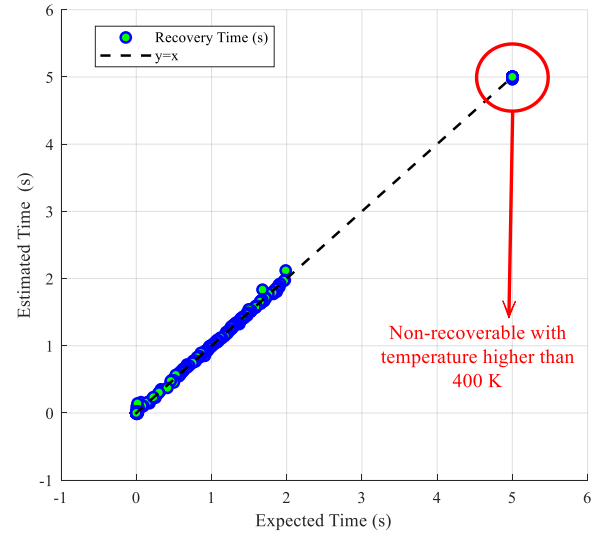


Fig. 17. The regression plot for the expected recovery time versus the estimated recovery time.

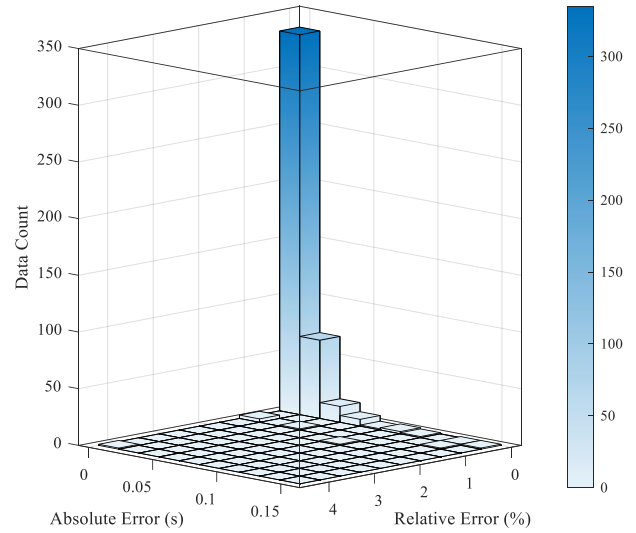


Fig. 18. Absolute error & relative error of estimation of the recovery time of SFCL.

are clustered around a very tiny absolute error and relative error. This suggests that in the proposed FFNN model, the estimated values are quite close to the expected values.

Fig. 15 shows the estimated superconducting layer temperature versus the expected temperature of the superconducting layer. Again, by considering $y=x$ line as an indication of accuracy, the estimated temperatures for the superconducting layer are in excellent coordination with expected values, which resulted in 0.9999 of the R-value.

Fig. 16 shows the absolute error and relative error of the temperature estimation for the superconducting layer. According to Fig. 16, absolute error is plotted along the x-axis, while relative error is presented as a percentage (%) on the y-axis. The z-axis indicates the data count for each error. The figure shows absolute error and relative error are close to zero for 99% of the data points while the rest of data points have the relative error between 0 to 1% and the absolute error less than 2 K.

Fig. 17 demonstrates the relationship between the expected and the estimated recovery times. The blue circles indicate individual recovery time observations, which fit the $y=x$ dashed line accurately, suggesting a strong correlation between the expected and estimated times. For the cases that the temperature is higher than 400 K, we consider the SFCL is non-recoverable. The absolute error and relative error of the recovery time are presented in Fig. 18. According to Fig. 18, most of the recovery time estimations have the relative error less than 1% and the absolute error less than 0.05 s, which indicates that the proposed FFNN model has high precision for estimating the recovery time of SFCL.

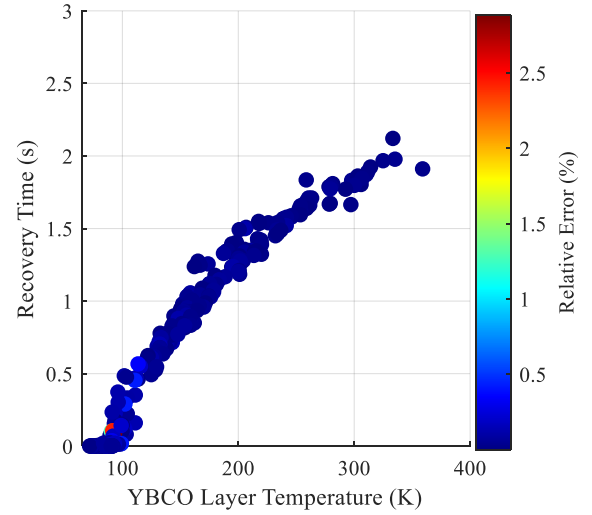
Fig. 19(a) shows the relative error of the temperature in superconducting layer of YBCO tapes versus recovery time. According to Fig. 19(a), recovery time increases when the temperature of the YBCO layer is increased. The relative error of the maximum limited fault current and the temperature of the superconducting layer is shown in Fig. 19(b). It can be found that the temperature of the superconducting layer is higher with a higher maximum fault current. Fig. 19(c) shows the relative error of the maximum resistance of SFCL and the temperature of the superconducting layer. According to the figure, the temperature of the superconducting layer increases while the maximum resistance of SFCL increases. More, all three figures show low relative error while 99.24% of the data have the relative error less than 0.5%, with maximum relative error 2.8%.

C. Results Comparison for different AI algorithms

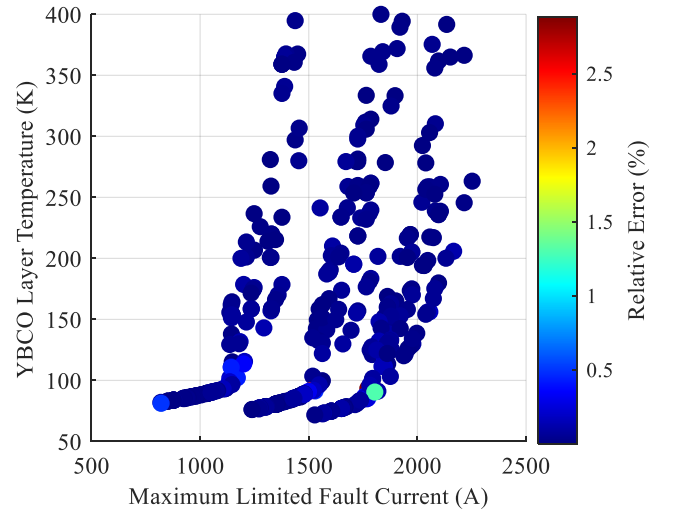
To compare the results of FFNN by other two artificial intelligent-based methods, a Bayesian optimization algorithm have been applied to GPR and GAM model, aiming to minimize their estimation error. The optimization algorithm has been conducted based on the procedure proposed by [62].

In this regard, the optimization minimizes the estimation error by changing the values of the hyperparameters of GPR and GAM models. The values of objective function during the optimization procedure are shown in Fig. 20. The GPR model, after 5 times of objective function evaluation, gets to its minimum value which is approximately equal to 4.5×10^{-3} while the minimum value of the GAM model is about 5.4×10^{-3} . By using the optimized hyperparameters of the GAM and GPR models, the result of their estimation is shown and compared in Table 5.

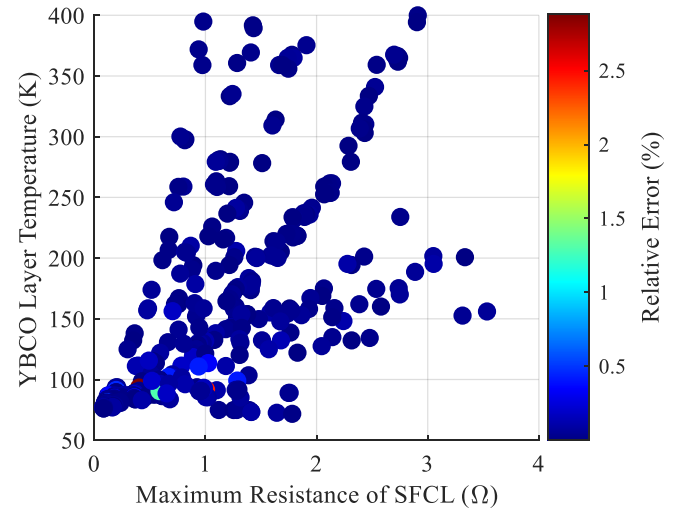
As shown in Table 5, lowest accuracy of the FFNN model is related to the estimation resistance with MAPE of 0.9049%, R value of 0.9988, and RMSE value of 0.0174. GPR model has 48% higher MAPE and about 12% higher RMSE value, compared with FFNN. The GAM model has 5 times higher MAPE and 4 times higher RMSE value, compared to FFNN model. As seen in Table 5, the performance of FFNN is better than two other methods, not only for estimation of SFCL resistance but also for estimation of all other outputs, including the limited current, fault energy, recovery time, and temperatures of different layers.



(a) Relative error of the YBCO Layer temperature & recovery time



(b) Relative error of the maximum limited fault current & YBCO Layer temperature



(c) Relative error of the maximum resistance of SFCL & YBCO Layer temperature

Fig. 19. Relative error of different inputs and outputs

TABLE V. MAPE, R, AND RMSE VALUES OF OPTIMIZED GPR AND GAM MODELS VERSUS FFNN RESULTS

method	FFNN			GPR			GAM		
Index	MAPE (%)	R	RMSE	MAPE (%)	R	RMSE	MAPE (%)	R	RMSE
Limited current	0.0761	0.9999	0.0001	0.0033	0.9999	4.7e-6	1.1637	0.9993	0.0016
Fault energy	0.0514	0.9999	146.727	0.3257	0.9999	1585.58	2.5293	0.9998	8315.3134
Resistance	0.9049	0.9998	0.0174	1.3422	0.9997	0.0191	5.7616	0.9988	0.04146
L-copper temperature	0.1805	0.9999	0.4129	0.7164	0.9999	1.5265	6.3415	0.9961	13.2268
U-copper temperature	0.1146	0.9999	0.2814	0.6944	0.9999	1.5281	6.3116	0.9956	15.4584
Hastelloy temperature	0.0777	0.9999	0.1891	0.7354	0.9999	1.5063	6.7147	0.9959	15.521
L-silver temperature	0.0886	0.9999	0.3656	0.6711	0.9999	1.3607	9.1472	0.9957	14.9834
U-silver temperature	0.2759	0.9999	1.0632	0.7151	0.9999	1.6059	6.5811	0.9948	17.0709
L-solder temperature	0.0729	0.9999	0.3467	0.7809	0.9999	1.7503	5.7958	0.9965	14.1009
U-solder temperature	0.3144	0.9999	0.7165	0.6355	0.9999	1.4072	6.4227	0.9965	13.4616
SC temperature	0.0645	0.9999	0.2238	0.6308	0.9999	1.4306	6.7116	0.9971	12.6825
L-substrate temperature	0.4041	0.9999	1.1045	0.6307	0.9999	1.2944	6.3817	0.9966	12.2358
U-substrate temperature	0.0673	0.9999	0.1538	0.6330	0.9999	1.6880	8.7127	0.9951	16.0788
Recovery time	0.0431	0.9999	0.0721	0.4762	0.9999	0.1074	8.6318	0.9497	0.4488

TABLE VI. NDD, AND RV, OF OPTIMIZED GPR AND GAM MODELS VERSUS FFNN RESULTS

Method	FFNN		GPR		GAM	
Index	NDD	RV	NDD	RV	NDD	RV
Limited current	0	1.1e-9	0	1.01e-9	4.4633	7e-4
Fault energy	0	1.05e-9	0	1.05e-7	17.7126	2e-4
Resistance	5.7208	2.5e-9	9.9429	4.5e-9	43.7714	2.2e-4
L-copper temperature	0.6865	1.3e-9	4.4571	3.2e-8	54.5143	3.9e-4
U-copper temperature	0.2288	1.2e-9	4.0000	2.3e-8	56.4571	4.4e-8
Hastelloy temperature	0	1.4e-9	3.5429	1.4e-9	55.2000	4.1e-4
L-silver temperature	0.2288	1.09e-9	4.1143	2.9e-8	53.6000	4.3e-4
U-silver temperature	0.4577	1.2e-9	4.5714	7.8e-8	55.5429	5.2e-4
L-solder temperature	0.2288	1.3e-9	5.1429	6.5e-8	57.9429	3.5e-4
U-solder temperature	0.9153	1.7e-9	2.6286	5.8e-7	55.1714	3.5e-4
SC temperature	0	1.6e-9	3.4286	7.7e-8	56.3429	2.9e-4
L-substrate temperature	1.3730	1.7e-9	2.0571	7.7e-8	58.2857	3.4e-4
U-substrate temperature	0	1.1e-9	2.7429	1.4e-7	58.9714	4.9e-4
Recovery time	3.2951	1.3e-9	3.8104	5.3e-8	8.1124	0.0503

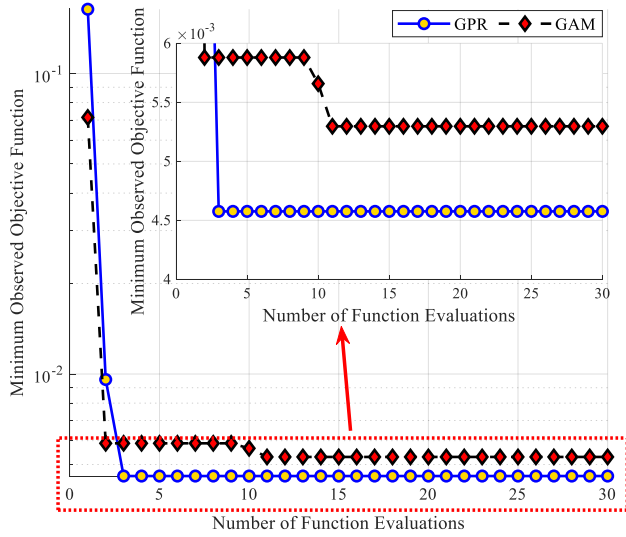


Fig. 20. Optimization procedure of GPR and GAM models with respect to the change of objective function versus number of objective functions evaluation

TABLE VII. TEST AND TRAIN TIME OF OPTIMIZED GPR AND GAM MODELS VERSUS FFNN RESULTS

Method	FFNN	GPR	GAM
Test time (s)	0.056	0.043	0.035
Train Time (s)	250.11	22.88	3.83

Table 6 shows the NDD and RV of different methods. NDD of the FFNN model. FFNN has better performance regarding the NDD while GPR and GAM models have high NDD in all target outputs. Meanwhile, average RV of the FFNN is about 10x to 100x lower than RV of GPR. Also, the RV of the FFNN is about 100,000x to 1,000,000x lower than RV of GAM which shows the highly accurate characteristic of FFNN for estimation of the outputs.

Finally, Table 7 tabulates test and train times for each of optimized methods. As seen in this table, test time of the FFNN is 30% higher than GPR and 60% higher than GAM method. On the other hand, the train time of FFNN is 12x higher than GPR and 80x higher than GAM. Although FFNN is a slower method, it has high accuracy compared to GPR and GAM.

V. CONCLUSION

Design, development, and condition monitoring of Superconducting Fault Current Limiters (SFCLs) is one of the most important challenges where temperature, resistance, and recovery time of the SFCLs are of interest. Currently, this is conducted through numerical models such as finite element methods, finite difference methods, and some semi-analytical methods such as Thermal Electrical Analogy (TEA) and equivalent circuit models. These simulation methods require a high amount of computational burden and long computational time which make them infeasible for real-time and system level modelling of sensitive applications such as electric aircraft. Some fast Artificial Intelligence (AI)-based methods need to be developed to model a SFCL during its operation, however; not to be constrained to specific fault scenarios under limited operational conditions.

To address this challenge, this paper proposed, developed and validated a novel model based on AI for estimating the

electrothermal characteristic of an SFCL, under different scenarios of faults. These scenarios include different fault duration, prospective fault current, voltages, initial operating temperature, fault energy, length of SFCL and the module number of SFCL. Feed Forward Neural Network (FFNN), Generalized Additive Model (GAM), and Gaussian Process Regression (GPR) are the AI based techniques used in this paper. The most important findings of the work could be summarized as follows:

- The FFNN model has the highest accuracy for the estimation of electrothermal characteristics of SFCL with average relative error of less than 1%.
- To support the above finding, it should be noted that the amount of data with accuracy less than 98% for FFNN is about 5%.
- The FFNN method has superiority comparing to GAM and GPR models with lower Mean Absolute Percentage Error (MAPE) and higher goodness of fit.
- In worst case scenario, the MAPE of the FFNN is approximately less than 0.2% while this number for GPR and GAM is 1.3% and 9.1%.
- Although FFNN performs better in terms of accuracy, the train/test times of the GPR and GAM models are way lower than FFNN.

Finally, it should be noted that the AI-based surrogate model is capable of estimating the behaviour of SFCLs under fault conditions by considering the variation of critical current density along the length of HTS tape. This study will be carried out in the future work.

ACKNOWLEDGEMENT

This work was supported by the U.K. Engineering and Physical Sciences Research Council (EPSRC) under Grant EP/X5257161/1.

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